

JECRC UNIVERSITY, JAIPUR
Session: Jan-June 2026
Subject: Mathematics-II (DMA025A)
B. Tech. II Semester I Year 2025-26
Tutorial Sheet-8 (M.M: 64)

UNIT-4: Complex Variable

CO4: The students will learn the complex variable- Differential.

PART – A (5*2=10)

A1: Define Analytical function.

A2: Define Harmonic function.

A3: State conjugate Harmonic function.

A4: Write Milne Thomson method.

A5: Write Mobius transformation.

PART – B (3*7=21)

B1: Write a short note on conformal mapping

B2: Determine whether $1/z$ is analytic or not.

B3: Find the bilinear transformation that maps the points $z_1 = \infty, z_2 = i$ and $z_3 = 0$ into the point $w_1 = \infty, w_2 = i$ and $w_3 = 0$

PART – C (3*11=33)

C1: Find the analytic function $f(z) = u(x, y) + iv(x, y)$ of which the real part $u(x, y) = e^x(x \cos y - y \sin y)$.

C2: Prove that $f(z) = \sinh z$ is analytic function of the complex variable $z = x + iy$.

C3. Find a mobius transformation of the upper half-plane $I(z) > 0$ on to the interior $|w| < 1$ of the unit circle $|w| = 1$
